AJ Rivas hux 960

# EE 4113 Homework 2 Assignment

**Homework 2 Overview:**

This assignment is worth 20 points. All *Homework 2* questions are based on the content available through the **Week 2 – Laplace Transforms** link listed in the EE 4113 Blackboard shell under **Content>Lectures**, and on the information below (Homework 2 Background).

**Homework 2 Background:**

Obtaining the ODE – Ordinary differential equations (ODEs) are formed to model circuit behavior by applying KVL and KCL. For simple RLC circuits the ODE can be formed using a single KVL or KCL expression through the use of the voltage-current relationships for each of the components:

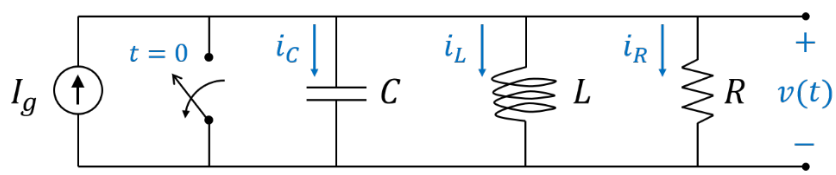
$$v_R(t) = i_R(t)R$$

$$i_C(t) = C\frac{dv_C(t)}{dt}$$

$$v_L(t) = L\frac{di_L(t)}{dt}$$

After forming the 'raw' KVL or KCL equation, modifications are typically made to form the ODE in terms of a single function.

Example – For example, given the following circuit,




the KCL equation is obtained to be,

$$i_R(t) + i_L(t) + i_c(t) = I_g$$

Then substituting in the voltage-current relationship we obtain (observe that $v_R = v_L = v_C = v(t)$),

$$\frac{v(t)}{R} + \frac{1}{L}\int_0^{\tau} v(t)\, d\tau + C\frac{dv(t)}{dt} = I_g$$

[For circuits without sources (i.e., if $I_g = 0$) we may simply differentiate (indirect approach) to remove the integral and form the ODE. Although, if there is a source we have to use a direct approach by making appropriate substitutions.]

We use the direct approach by observing that by differentiating $v(t) = L\frac{di_L(t)}{dt}$ we obtain

$$\frac{dv(t)}{dt} = L\frac{di_L^2(t)}{dt^2}$$

Hence, we may choose to express the KCL expression as follows (keeping the $i_L(t)$ term and performing two substitutions associated with $i_R(t)$ and $i_C(t)$),

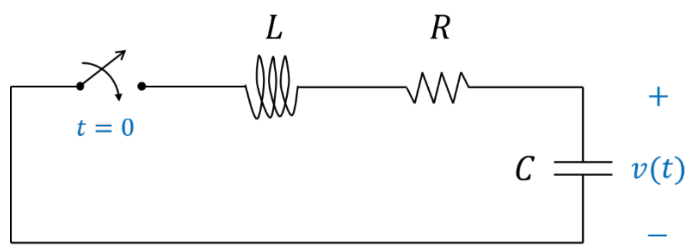
$$\frac{\left(L\frac{di_L(t)}{dt}\right)}{R} + i_L(t) + C\left(L\frac{di_L^2(t)}{dt^2}\right) = I_g$$

Simplifying this expression gives us the appropriate form for the ODE:

$$\frac{di_L^2(t)}{dt^2} + \frac{1}{RC}\frac{di_L(t)}{dt} + \frac{1}{LC}i_L(t) = \frac{I_g}{LC}$$

**Homework 2 Questions:**

For the given circuit:




1. (2 points) Determine the KVL or KCL equation needed to model the circuit behavior.

2. (2 points) Show the steps needed to model the circuit as an ordinary differential equation (ODE) with constant coefficients in terms of the function $i(t)$.

   Hint: you may use indirect approach and simply differentiate both sides of the expression obtained in 1.

3. (4 points) Assuming that the initial conditions $i(0)$ and $i'(0)$ are non-zero, transform the time-domain ODE with constant coefficients into the s-domain via the Laplace Transform. Present the Laplace version with the format: $I(s) = \frac{m(s)}{q(s)} + \frac{p(s)}{q(s)}R(s)$

   Hint: The input is $R(s) = 0$, so there will only be the term $\frac{m(s)}{q(s)}$

*Continue to next page…*

4. (8 points) Assume the values for the circuit components are $R = 560\Omega$, $L = 100mH$ and $C = 0.1\mu F$. Perform partial fraction decomposition on the expression obtained in 3. Solve and obtain the particular values for the partial terms using the initial condition information $i(0) = 0\ A$, and $i'(0) = 1000\ A/s$.

   Hint: Given $s^2 + as + b = 0$ the roots may be expressed as

   $$s_{1,2} = \frac{-a \pm \sqrt{a^2 - 4b}}{2}$$

   Hint: If the roots are complex then the partial fraction decomposition will require you to match one or more of the available Laplace inverse transforms (you should obtain and review a Laplace transforms table). In particular this problem will use the inverse transforms associated with the forms,

   $$\frac{(s-a)}{(s-a)^2 + b^2}$$

   and

   $$\frac{b}{(s-a)^2 + b^2}$$

   In order for you to do this, you will need to use the following expansion:

   $$\frac{m(s)}{(s-a)^2 + b^2} = A\left[\frac{(s-a)}{(s-a)^2 + b^2}\right] + B\left[\frac{b}{(s-a)^2 + b^2}\right]$$

   The denominator $(s-a)^2 + b^2$ is a form of the original complex polynomial $s^2 + \frac{R}{L}s + \frac{1}{LC}$ obtained by 'completing the square' (you may have to review/look-up the topic 'completing the square' to complete this part).

5. (4 points) Given the values $R = 560\Omega$, $L = 100mH$ and $C = 0.1\mu F$ simulate the response $i(t)$ using PSpice assuming information $i_L(0) = 0\ A$ and $v_C(0) = 100V$?

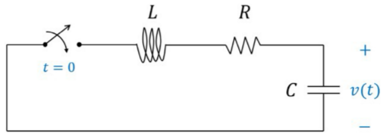
   Hint: Insert initial condition information by selecting part (e.g., double click on the part) and opening its *properties* page. Place information in *properties* page under the *IC* field. For a capacitor this will be a voltage, for the inductor it will be a current.

   Hint: Use a 3 $ms$ in the 'Run to time' field in the *Simulation Settings* window.

   Hint: You are measuring current, so you will need to use a current probe.

**Homework 2 Questions:**

For the given circuit:




1. (2 points) Determine the KVL or KCL equation needed to model the circuit behavior.

2. (2 points) Show the steps needed to model the circuit as an ordinary differential equation (ODE) with constant coefficients in terms of the function $i(t)$.

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   Hint: The input is $R(s) = 0$, so there will only be the term $\frac{m(s)}{q(s)}$

1) KVL

$$V_L + V_R + V_C = 0$$

$$\Rightarrow L\frac{di(t)}{dt} + Ri(t) + \frac{1}{C}\int i(t)\,dt = 0$$

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2) Convert to ODE

$$\frac{1}{dt}\left[L\frac{di(t)}{dt} + Ri(t) + \frac{1}{C}\cancel{\int} i(t)\,\cancel{dt} = 0\right]\frac{1}{dt}$$

$$= L\frac{d^2 i(t)}{dt^2} + R\frac{di(t)}{dt} + \frac{1}{C}i(t) = 0$$

2nd order ODE

3) Laplace Transform

Time domain
(t) → s
laplace

$$Ls^2 I(s) + Rs\,I(s) + \frac{1}{C}I(s) = 0$$

$$L(s^2 I(s) - si(0) - i'(0)) + R(sI(s) - i(0)) + \frac{1}{C}I(s) = 0$$

$$I(s)\left[Ls^2 + Rs + \frac{1}{C}\right] = Lsi(0) + Li'(0) + Ri(0)$$

$$I(s) = \frac{Lsi(0) + Li'(0) + Ri(0)}{Ls^2 + Rs + 1/C} \begin{matrix} \leftarrow m(s) \\ \leftarrow q(s) \end{matrix}$$

$$I(s) = \frac{m(s)}{q(s)} + \boxed{\cancel{\frac{p(s)}{q(s)} \cdot R(s)}}\ 0$$

$R(s) = 0$

$$I(s) = \frac{m(s)}{q(s)}$$

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4. (8 points) Assume the values for the circuit components are $R = 560\Omega$, $L = 100mH$ and $C = 0.1\mu F$. Perform partial fraction decomposition on the expression obtained in 3. Solve and obtain the particular values for the partial terms using the initial condition information $i(0) = 0\ A$, and $i'(0) = 1000\ A/s$.

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   The denominator $(s-a)^2 + b^2$ is a form of the original complex polynomial $s^2 + \frac{R}{L}s + \frac{1}{LC}$ obtained by 'completing the square' (you may have to review/look-up the topic 'completing the square' to complete this part).

$$I(s) = \frac{Lsi(0) + Li'(0) + Ri(0)}{Ls^2 + Rs + 1/C}$$

1000 A/s

0

0A

Given Values → sub

$L = 100mH \rightarrow 0.1\,H$

$R = 560\Omega$

$C = 0.1\mu F \rightarrow 10^{-7}$

$$I(s) = \frac{(0.1\cancel{s \cdot 0}) + (0.1 \cdot 1000) + (5\cancel{60 \cdot} 0)}{0.1s^2 + 560s + 1/10^{-7}}$$

0 100 0

$$\Rightarrow \frac{100}{0.1s^2 + 560s + 10^7} \times 0.1 \Rightarrow \frac{1000}{s^2 + 5600s + 10^8}$$

$$s^2 + \underset{a}{5600s} + \underset{b}{10^8} = 0$$

$$s_{1,2} = \frac{-a \pm \sqrt{a^2-4b}}{2} = \frac{-(5600) \pm \sqrt{(5600)^2 - 4(10^8)}}{2} \Rightarrow \frac{-5600 \pm \sqrt{31360000 - 400000000}}{2}$$

$$\Rightarrow \frac{-5600 \pm \sqrt{-368640000}}{2} \Rightarrow \frac{-5600 \pm j\sqrt{368640000}}{2} \Rightarrow \frac{-5600 \pm j\,19200}{2}$$

$$s_{1,2} \doteq \underset{a}{-2800} \pm j\,\underset{b}{9600} \quad \left[ s = a + jb \right] \quad \left\{ \frac{s-a}{(s-a)^2+b^2},\ \frac{b}{(s-a)^2+b^2} \right\} \quad (s+2800)^2 + (9600)^2$$

$$\Rightarrow \quad A = 1000 \quad \frac{A}{(s+2800)^2+(9600)^2} \Rightarrow I(s) = A\left[\frac{s-(-2800)}{(s+2800)^2+(9600)^2} + B\,\frac{9600}{(s+2800)^2+(9600)^2}\right]$$

$$i(0) = 0 \quad A\cancel{e^0}^{1}\cancel{\cos(0)}^{1} + B\cancel{e^0}^{1}\cancel{\sin(0)}^{0} \Rightarrow \underline{A = 0}$$

$$i'(t) = A\left[-2800e^{-2800t}\cos(9600t) - 9600e^{-2800t}\sin(9600t)\right] +$$
$$B\left[-2800e^{-2800t}\sin(9600t) + 9600e^{-2800t}\cos(9600t)\right]$$

$$\Rightarrow Ae^{-2800t}\left[-2800\cos(9600t) - 9600\sin(9600t)\right] +$$
$$Be^{-2800t}\left[-2800\sin(9600t) + 9600\cos(9600t)\right]$$

$$i'(0) = 1000 = Ae^0\left[-2800\cos(0) - 9600\cancel{\sin(0)}^{0}\right] + Be^0\left[-2800\cancel{\sin(0)}^{0} + 9600\cos(0)\right]$$

$$\Rightarrow 1000 = -2800A + 9600B$$

$$\rightarrow 100 = -2800(0) + 9600B$$

$$B = \frac{1000}{9600} = \frac{1}{9.6}$$

$$\boxed{i(t) = \frac{1}{9.6}e^{-2800t}\sin(9600t)}$$

5. (4 points) Given the values $R = 560\Omega$, $L = 100mH$ and $C = 0.1\mu F$ simulate the response $i(t)$ using PSpice assuming information $i_L(0) = 0\ A$ and $v_C(0) = 100V$?

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